

Colgan Air Flight 3407

Flight model: Bombardier Q400

Date: February 12, 2009

From: Newark, USA

To: Buffalo, USA

The Bombardier Q400 is a twin engine turbo prop. The first officer was a trainee in the flight

Problem

The weather is seen to be rainy. While preparing for landing, when the landing gears are deployed and flaps extended which is done to provide more lift to slow the plane down. The control column starts shaking. The plane starts tilting towards the left and starts to fall. The flight becomes uncontrollable at this movement. The it roll towards the right and keeps falling.

Investigation

In order to find if the jet was fully integrated during crash. The investigators look for the 4 corners nose, tail and wing tip of either side of the wing. In the DVR, it is found that right before landing the crew noticed build up of ice around the wings of the plane. Ice increases the air drag thereby requiring more power to maintain the airspeed. The crew reports no de-icing procedure performed. Minutes later, they start experiencing stick shaker warning indicating a stall. But, the plane like this had systems to prevent ice buildup. There is a rubber bladder in the front side of the wing Air is pumped into these bladders causing it to expand and break the ice. The FDR, shows the crew had turned on the de-icing equipment. The power required to fly and speed from the FDR brings to the conclusion that the ice build up was normal within safe limits. At that flap setting, the stick shaker warning should have been triggered at 111 knots but here it had been triggered at 131 knots. This is because, this specific plane has something called as reference speed switch which controls the sensitivity of the stall warning. Pilots should be turning on the switch when flying through icing conditions. The pilots did turn it on, but did not increase the air speed to required levels during icing condition despite warning in the ADI showing a stall warning maybe triggered. It is found that the fight had not been into stall till the stick shaker warning was triggered. When the plane stalled the pilot must have pitched the nose forward to gain speed which is normal response to a stick shaker rather the captain pulled the nose up this leads to reducing the airspeed. The first officer had retracted the flaps that reduced the lift, making the plane drop faster. The investigator finds that the pilots were talking with each other during the flight and even during the landing approach. This is a violation of the rule of **sterile cockpit**, which mandate that the crew should not talk anything other than conversation essential for the flight during critical phases of the flight here it is during the landing phase. The distracted crew oversaw the ADI warning. The crew also shows signs of fatigue. The pilots had to travel a lot to take this flight there they did not get enough rest.

Changes that were brought after the incident

The senate instructed FAA had put up laws on pilot fatigue.

Reference

1. The Worst Plane Crash Of 2009 — Flight 3407 — Mayday: Air Disaster
<https://www.youtube.com/watch?v=U6tCJteMHRk>